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भारतीय मानक मसौदा विवो अनुप्रयोगों में प्रत्यारोपण योग्य उपकरणों के लिए उत्पादों के एडिटिव विनिर्माण पर मार्गदर्शन

Guidance on Additive Manufacturing of Products for in vivo Applications and Implantable Devices

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FOREWORD

Additive Manufacturing (AM), is a manufacturing process of joining materials to make parts from 3D model data, usually layer upon layer. In the field of medical devices, AM enables the fabrication of complex geometries, controlled porosity, lattice structures, internal channels, and patient-specific implants that are not readily achievable using conventional subtractive or formative manufacturing methods.

For in vivo applications and implantable devices, the performance of additively manufactured products is influenced not only by the base material composition but also by feedstock characteristics, build orientation, layer thickness, energy input, thermal history, post-processing treatments, and surface finishing operations. The layer-wise fabrication mechanism can result in microstructural anisotropy, residual stresses, partially fused particles, internal porosity, and surface irregularities. These characteristics may significantly affect mechanical strength, fatigue resistance, corrosion behaviour, tribocorrosion performance, biological response, and long-term functional stability of implants.

In addition, the digital workflow-from medical image acquisition (e.g., CT or MRI), segmentation, computer-aided design (CAD), data conversion (e.g., STL or AMF), slicing, and generation of machine-executable instructions-introduces specific considerations related to dimensional fidelity, geometric accuracy, coordinate integrity, and reproducibility. Errors in data processing or conversion may propagate through the manufacturing chain and influence final implant performance.

Feedstock materials used in AM, including metallic powders, polymeric filaments, resins, gels, and composite biomaterial inks, require careful characterization with respect to particle size distribution, morphology, flowability, rheological behaviour, thermal transitions, crystallinity, chemical composition, and potential contamination. Material transformations during processing, such as oxidation, phase transformation, crosslinking, or degradation, can alter the physicochemical and mechanical properties of the final implant. Where multiple materials are combined, interfacial compatibility and interaction effects require additional evaluation.

Post-processing operations such as stress relieving, annealing, hot isostatic pressing, UV curing, machining, abrasive finishing, chemical treatment, cleaning, and sterilization can substantially modify surface topography, microstructure, residual stress distribution, and material properties. These processes therefore require validation and verification to ensure that intended design characteristics and functional performance are achieved without introducing unintended defects or contaminants

This standard provides technical guidance on:

- feedstock material qualification and lifecycle considerations;
- digital data acquisition, processing, and CAD-to-machine workflow;
- design considerations specific to additive manufacturing, including lattice structures and minimum feature size;
- process validation and control of AM systems;
- post-processing, cleanliness, and sterilization considerations;
- physico-chemical, mechanical, microstructural, and biological evaluation of AM implants;
- test methods relevant to dimensional accuracy, porosity assessment, corrosion behaviour, tribocorrosion, mechanical performance, and structural integrity;
- risk analysis aspects specific to additively manufactured implantable devices.

This document is intended to serve as a comprehensive technical framework for the design, manufacture, post-processing, and evaluation of products manufactured by additive manufacturing for in vivo applications and implantable use. It addresses considerations that are unique to or require additional attention in the context of layer-wise manufacturing technologies.

1 SCOPE

This document provides guidance on biomaterials used in additive manufacturing (AM) technology to build up Implantable physical 3D Geometries by successive addition of material.

The implantable geometries can be classified based on the specific field in medical application

The document provides and guidance on procedures in the process and evaluation of 3D printed implants potential application as medical device and to avoid to cause adverse reaction.

This document provides guidance regarding Physico-chemical evaluation and biological evaluation of implants made by AM technology to bring awareness of critical factors and reduce generation of erroneous or misleading test results that could pose potential risk to health.

2 NORMATIVE REFERENCE

The documents referenced below are indispensable for the application of this standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document applies

All cross reference will be listed here.

IS No.	IS Title
IS 19491:2026	Implants for Surgery Test Solutions and Environmental Conditions for Static and Dynamic Corrosion Tests on Implantable Materials and Medical Devices
IS 1586 :2025	Metallic materials Rockwell hardness testPart 1: Test method (Sixth Revision)
IS 19227 :2025	Breathing System Filters for Anaesthetic and Respiratory Use Part 1: Salt Test Method to Assess Filtration Performance
IS 17932 :2024	Biological Evaluation of Medical Devices Part 2 Framework for identification and quantification of potential degradation products
IS 17932 :2023	Biological evaluation of medical devices Part 1: Evaluation and testing within a risk management process (ISO 10993-1 : 2018, MOD)
IS 18076 :2023	Non-active surgical implants - General requirements
IS/ISO 80000 :2022	Quantities and Units Part 1 General
IS/ISO 14971 :2021	Medical devices - Application of risk management to medical devices First Revision
IS/ISO 10993 :2021	Biological Evaluation of Medical Devices Part 12 Sample Preparation and Reference Materials
IS/ISO 19227 :2018	Implants for Surgery - Cleanliness of Orthopedic Implants - General Requirements
IS/ISO 13485 :2016	Medical Devices — Quality Management Systems — Requirements for Regulatory Purposes (First Revision)
IS 13360 :1992	Plastics - Methods of testing: Part 5 mechanical properties section 13 determination of rockwell hardness
IS 3073 :1967	Assessment of surface roughness

3 TERMS AND DEFINITIONS

For the purposes of this document, the following terms and definitions shall apply.

3.1 General Terms

3.1.1 3D printer

Machine used for 3D printing

3.1.2 Additive Manufacturing

Process of joining materials to make parts from 3D model data, usually layer upon layer, as opposed to subtractive manufacturing and formative manufacturing methodologies

3.1.3 Additive Manufacturing System

additive manufacturing equipment

3.1.4 AM Machine

Section of the additive manufacturing system including hardware, machine control software, required set-up software and peripheral accessories necessary to complete a build cycle for producing parts

3.1.5 Multi-Step Process

Type of additive manufacturing process in which parts are fabricated in two or more operations where the first typically provides the basic geometric shape and the following consolidates the part to the fundamental properties of the intended material

3.1.6 Single-step process

Type of additive manufacturing process in which parts are fabricated in a single operation where the basic geometric shape and basic material properties of the intended product are achieved simultaneously

3.2 Process categories

3.2.1 Binder Jetting

Additive manufacturing process in which a liquid bonding agent is selectively deposited to join powder materials

3.2.2 Directed Energy Deposition

Additive manufacturing process in which focused thermal energy is used to fuse materials by melting as they are being deposited

NOTE :

Focused thermal energy” means that an energy source (for example laser, electron beam or plasma arc) is focused to melt the materials being deposited.

3.2.3 Material Extrusion

Additive manufacturing process in which material is selectively dispensed through a nozzle or orifice

3.2.4 Material Jetting

Additive manufacturing process in which droplets of feedstock material are selectively deposited

3.2.5 Powder Bed Fusion

Additive manufacturing process in which thermal energy selectively fuses regions of a powder bed

3.2.6 Vat Photopolymerization

Additive manufacturing process in which liquid photopolymer in a vat is selectively cured by light-activated polymerization

3.3 Processing: General

3.3.1 3D Printing

Fabrication of objects through the deposition of a material using a print head, nozzle or another printer technology

3.3.2 Build Chamber

Enclosed location within the additive manufacturing system where the parts are fabricated

3.3.3 Build Cycle

Single process cycle in which one or more components are built by successive joining of material within the build space of the additive manufacturing system

3.3.4 Build Platform

<of a machine> Base which provides a surface upon which the building of the part/s, is started and supported throughout the build process

3.3.5 Build Space

Location where it is possible for parts to be fabricated, typically within the build chamber or on a build platform

3.3.6 Build Surface

Area where material is added, normally on the last deposited layer, which becomes the foundation upon which the next layer is formed

NOTES:

1. For the first layer, the build surface is often the build platform.
2. In the case of directed energy deposition processes, the build surface can be an existing part onto which material is added.

3.3.7 Build Volume

Total usable volume available in the machine for building parts

3.3.8 Feed Region

In powder bed fusion location/s in the machine where feedstock is stored and from which a portion of the feedstock is repeatedly conveyed to the powder bed during the build cycle

3.3.9 Machine Coordinate System, noun

Three-dimensional coordinate system as defined by a fixed point on the build platform with the three principal axes labelled x-, y-, and z-, with rotary axis about each of these axis labelled A, B, and C, respectively, where the angles between x-, y- and z- can be Cartesian or defined by the machine manufacturer

NOTE :

Machine coordinate system is fixed relative to the machine, as opposed to coordinate systems associated with the build surface which can be translated or rotated.

3.3.10 Manufacturing Lot

Set of manufactured parts having commonality between feedstock, production run, additive manufacturing system and post-processing steps (if required) as recorded on a single manufacturing work order.

NOTE :

The additive manufacturing system can include one or several AM machines and/or post-processing machine units as agreed by AM provider and customer.

3.3.11 Manufacturing Plan

Document setting out the specific manufacturing practices, technical resources and sequences of activities relevant to the production of a particular product including any specified acceptance criteria at each stage

3.3.12 Process Chain

Sequence of operations necessary for the part to achieve desired functionality and properties

3.3.13 Process Chain

Sequence of operations necessary for the part to achieve desired functionality and properties

3.4 Processing: Data

3.4.1 Additive Manufacturing File Format

File format for communicating additive manufacturing model data including a description of the 3D surface geometry with native support for colour, materials, lattices, textures, constellations and metadata

NOTE :

Additive Manufacturing File Format (AMF) can represent one of multiple objects arranged in a constellation. Similar to STL, the surface geometry is represented by a triangular mesh, but in AMF the triangles can also be curved. AMF can also specify the material and colour of each volume and the colour of each triangle in the mesh.

3.4.2 Product Data Exchange Specification (PDES)

Data exchange specification using STEP

3.4.3 Step

Standard for the exchange of product model data

NOTES :

1. This Standard that provides a representation of product information along with the necessary mechanisms and definitions to enable product data to be exchanged. The representation of product information, including components and assemblies, the exchange of product data, including storing, transferring, accessing and archiving.
2. commonly referred to as STEP-NC, specifies the slicing operation and other mechanical commands in the AM process.

3.4.4 STL

File format for model data describing the surface geometry of an object as a tessellation of triangles used to communicate 3D geometries to machines in order to build physical parts

NOTE :

The STL file format was originally developed as part of the CAD package for the early STereoLithography Apparatus, thus referring to that process.

3.5 Processing: Material**3.5.1 Batch**

<of feedstock> Defined quantity of feedstock with uniform properties and composition

NOTES :

1. One batch of any feedstock can be used in one or more production runs using different process parameters.
2. For some types of feedstock, for example powders and resins, one batch can consist of virgin material, used material or a blend of virgin and used materials.

3.5.2 Feedstock

bulk raw material supplied to the additive manufacturing building process

3.5.3 laser sintering

powder bed fusion process used to produce objects from powdered materials using one or more lasers to selectively fuse or melt the particles at the surface, layer upon layer, in an enclosed chamber

3.5.4 Lot

<of feedstock> quantity of feedstock produced under traceable controlled conditions from a single manufacturing process cycle

NOTES :

1. The size of a feedstock lot is determined by the feedstock supplier. It is common that the supplier distributes a portion of a lot to different AM system users.
2. Source documentation of the feedstock lot is required for several AM product applications. Source documentation is referred to as a "statement of conformity" and an "inspection document", or sometimes also referred to as a "certificate of conformance", "factory certificate" or "certificate of analysis".

3.6 processing: Material Extrusion**3.6.1 Filament**

feedstock characterized by extreme length relative to its uniform cross section

NOTES :

1. The size of a feedstock lot is determined by the feedstock supplier. It is common that the supplier distributes a portion of a lot to different AM system users.
2. Filaments made of metal are commonly referred to as wire.

3.6.2 Pellets

small mass of preformed feedstock material, having relatively uniform dimensions in any given batch

3.6.3 Prototype

physical representation of all or a component of a product that, although limited in some way, can be used for analysis, design and evaluation

3.7 Other Terms

3.7.1 *Digital Model Data*

The source data used to build the 3D geometry of the implant

3.7.2 *CAD Model Generation*

The process of creating the digital representation of the implant, which must follow principles to ensure accuracy before being converted to G-Code.

3.7.3 *Data Integrity and Traceability*

The standard defines the requirements for maintaining the security and accuracy of digital information throughout the conversion and transmission process

3.7.4 *Standardized Implants*

Off-the-shelf AM parts

3.7.5 *Patient-matched Implants*

Derived from medical imaging data

3.7.6 *Patient-specific Implants*

Uniquely designed for individual anatomical conditions

3.7.7 *Implantable Device*

A medical device intended to remain in the body for therapeutic or diagnostic purposes

3.7.8

Biocompatibility

Ability of a material to perform with an appropriate host response in a specific application

3.7.9

Porosity

Measure of void spaces in material, critical for Osseointegration

3.7.10 *Surface Model*

Mathematical or digital representation of an object as a set of planar or curved surfaces, or both, that can, but does not necessarily have to represent a closed volume

4 IMPLANTS BY ADDITIVE MANUFACTURING

4.1 General

4.1.1

Additively manufactured implants are implantable medical devices or components fabricated using layer-wise material deposition or consolidation processes based on digital model data. Such implants may be:

- a) standardized implants;
- b) patient-matched implants derived from medical imaging data;
- c) patient-specific implants uniquely designed for an individual anatomical condition.

4.2

Additive Manufacturing Process Categories

4.2.1

Implants covered by this document may be produced using additive manufacturing process categories. These include, but are not limited to:

- a) Vat photopolymerization, including Stereolithography (SLA)
- b) Powder bed fusion (PBF), including Selective laser sintering (SLS), Selective laser melting (SLM), Electron beam melting (EBM)
- c) Binder jetting
- d) Directed energy deposition (DED)

- e) Sheet lamination
- f) Material extrusion, including Fused filament fabrication (FFF)
- g) Material jetting, including PolyJet systems.

4.2.2

The applicable process category and sub-technology used for manufacture of the implant shall be clearly identified and documented.

4.3 Implant-specific Considerations

4.3.1

Additively manufactured implants may incorporate features such as:

- a) controlled porosity or lattice structures;
- b) internal channels or cavities;
- c) functionally graded regions;
- d) surface textures designed to promote osseointegration;
- e) customized geometries derived from patient imaging data.

4.3.2

The influence of process parameters, build orientation, and thermal history on these features shall be evaluated.

4.4 Performance implications

4.4.1

The mechanical, microstructural, and surface properties of additively manufactured implants can differ from those of conventionally manufactured implants of identical nominal composition. Differences may arise from:

- a) rapid solidification and cooling rates;
- b) layer-wise bonding mechanisms;
- c) residual stress development;
- d) partial fusion or entrapped feedstock;
- e) surface stair-stepping effects.

These differences shall be considered during design verification and performance evaluation.

4.5 Biological and safety evaluation

4.5.1

Additively manufactured implants shall be evaluated for biological safety and performance in accordance with applicable standards.

4.5.2

Biological evaluation shall be conducted using a risk management approach in accordance with , taking in to consideration: [IS 17932](#)

- a) the nature of body contact;
- b) the duration of exposure;
- c) the material composition and potential process-related changes;
- d) residual materials arising from additive manufacturing and post-processing;
- e) degradation products, where applicable.

4.5.3

Where process-specific characteristics such as porosity, lattice architecture, residual feedstock, or surface modifications may influence biological response, additional evaluations shall be performed as appropriate.

5 FEEDSTOCK MATERIALS

5.1 General

5.1.1 Feedstock materials used for additive manufacturing of implants shall be selected and qualified to ensure that the final manufactured product meets specified mechanical, physicochemical, and biological requirements.

5.1.2

Feedstock shall be considered a critical input material whose properties directly influence the microstructure, crystallinity, porosity, surface morphology, and functional performance of the finished implant.

5.2 Material Categories

5.2.1

Feedstock materials may include, but are not limited to:

- a) metallic powders;
- b) polymeric filaments;
- c) photopolymer resins;
- d) ceramic powders;
- e) hydrogels, gels, pastes, or biomaterial inks;
- f) composite or multi-material systems.

5.2.2

Where multiple materials are used in a single build, individual and combined material performance shall be evaluated.

5.3 Physicochemical characterization

5.3.1

Feedstock shall be characterized, as applicable, for the following properties:

- a) particle size distribution and morphology (for powders);
- b) flowability and apparent density;
- c) chemical composition and impurity levels;
- d) moisture content;
- e) rheological behaviour (for inks, gels, or resins);
- f) thermal transitions (e.g., melting temperature, glass transition temperature);
- g) degree of crystallinity;
- h) residual solvent or monomer content.

5.3.2 Characterization methods shall be appropriate to the feedstock type and additive manufacturing process.

5.4 Degree of crystallinity

5.4.1

Where applicable (e.g., semi-crystalline polymers, composite systems), the degree of crystallinity of the feedstock and printed construct shall be determined using Differential Scanning Calorimetry (DSC) or equivalent thermal analysis methods. The evaluation shall consider:

- a) melting enthalpy;
- b) crystallization behaviour;
- c) thermal transitions;
- d) changes before and after printing;
- e) effects of post-processing.

5.4.2

The degree of crystallinity shall be calculated using the enthalpy of fusion relative to that of a fully crystalline reference material, where applicable.

5.4.3

Changes in crystallinity due to additive manufacturing processing or post-processing shall be assessed for potential impact on mechanical and biological performance. Degree of crystallinity (using differential scanning calorimetry) is:

- a) 3D print the biomaterial inks
- b) DSC has to be done with two heating cycles from 40 °C to 250 °C, at a rate of
- c) 10 °C/min, to evaluate the thermal phenomena of the materials.
- d) The degree of crystallinity of build material or the biomaterial ink may be evaluated with Eq. below,

$$X_c = (\Delta H_f / (\Delta H_{f100} \times W_f)) \times 100$$

where X_c is the degree of crystallinity, ΔH_f is the fusion enthalpy obtained on other heating, and ΔH_{f100} is the fusion enthalpy for 100% crystalline material (biomaterial ink) and W_f represents the mass fraction of additive in biomaterial ink. J/g

- e) Use licensed software to obtain the graphical presentations, if required.

5.5 Chemical composition testing

5.5.1

The chemical composition of feedstock materials and printed constructs shall be verified to ensure consistency, purity, and absence of unintended contaminants. Testing may include, as appropriate:

- a) elemental analysis;
- b) spectroscopic methods;
- c) chromatographic methods;
- d) X-ray-based techniques.

5.5.2

Chemical composition testing shall confirm:

- a) conformity to specified formulation;
- b) absence of harmful impurities;
- c) stability of formulation after processing.

5.6 Chemical Characterization of Biomaterial ink

5.6.1

For biomaterial inks, hydrogels, or resin systems, chemical characterization shall be performed on:

- a) raw materials;
- b) formulated feedstock;
- c) printed construct (where relevant).

5.6.2

Characterization may include:

- a) Fourier Transform Infrared Spectroscopy (FTIR) or Attenuated Total Reflectance FTIR (ATR-FTIR);
- b) Proton Nuclear Magnetic Resonance (¹H NMR) spectroscopy;
- c) X-ray diffraction (XRD) for crystalline phase analysis;
- d) verification of crosslinking or polymerization state.

5.6.3

Where solvents or reactive components are used, residual solvent or unreacted monomer content shall be determined using suitable analytical techniques (e.g., gas chromatography).

5.6.4 Chemical characterization shall ensure that additive manufacturing and post-processing do not introduce undesirable chemical alterations.

5.7 Material Transformation During Processing

5.7.1

Material properties may change during additive manufacturing due to:

- a) thermal cycling;
- b) oxidation or gas uptake;
- c) crosslinking or curing reactions;
- d) degradation;
- e) phase transformation.

5.8 Feedstock Storage, Handling and Reuse

5.8.1

Feedstock storage and handling conditions shall be controlled and documented to prevent contamination or degradation. Where reuse of feedstock is permitted:

- a) reuse limits shall be defined;
- b) degradation effects shall be evaluated;
- c) traceability of reused material shall be maintained.

5.9 Biological Evaluation of Materials

5.9.1

Biological evaluation of feedstock materials and processed constructs shall be performed in accordance with IS 17932-1, using a risk-based approach that considers potential process-induced chemical or structural changes.

6 DESIGN:

6.1

The design of additively manufactured implants shall ensure that the finished device meets specified functional, mechanical, physicochemical, and biological performance requirements. Design shall consider:

- a) intended clinical application;
- b) anatomical constraints;
- c) loading conditions;
- d) material properties;
- e) additive manufacturing process limitations;
- f) post-processing requirements;
- g) process-induced anisotropy;
- h) dimensional variability inherent to additive manufacturing.

6.2

Design activities should take into account:

- a) The opportunities and design freedoms that AM offers designers
- b) The issues that designers should consider when designing parts for AM
- c) Warnings to designers, or red flag issues, that indicate situations that often lead to problems in many AM systems.

6.3 Design for Additive Manufacturing (DfAM)**6.3.1**

Design shall account for process-specific capabilities and limitations, including:

- a) minimum feature size relative to machine resolution;
- b) overhang limitations;
- c) support structure requirements and removability;
- d) achievable surface finish;
- e) thermal distortion and shrinkage;
- f) build orientation effects;
- g) layer bonding characteristics.

6.3.2

The resolution of the additive manufacturing system shall be considered to avoid mismatch between design intent and machine capability, which may result in unintended geometric artefacts or porosity.

6.3.3

Design modifications introduced to ensure manufacturability shall be verified for performance impact.

6.4 Design Controls**6.4.1 Computer-Aided Design (CAD) Models**

- a. Designs for additive manufacturing shall follow the principles of functional integration, minimum feature size, and support structure minimization.
- b. High quality and good resolution 3D medical images produced by scanning technologies such as CT, MRI and 3D scanners.

6.4.2 Customization & Personalization

For patient-specific devices, patient anatomy data (e.g., MRI, CT scans) must be incorporated into the CAD model in a secure and accurate manner.

6.4.3 Post-processing of images

- a) The post processing of images of DICOM (Digital Imaging and Communications in Medicine) may be performed to suit to the clinical use of the implant or patient specific and date rebuild to create CAD model.
- b) A processing software can be used to transfer CAD data into Standard Triangulation Language (STL) format
- c) The STL format data file serves as the input for the layering or slicing process in the same or another software to create Geometric Coordinates (G-Codes) which is sent to 3D printing machine for implant production.

6.5 Data processing:

This subclause is mostly relevant for custom-made and patient-matched implants or accessories.

6.5.1 Medical image data is often necessary to customize these devices. Optimized medical image data shall be used to minimize risks associated with data processing during additive manufacturing of implants.

6.5.2 The following factors related to data processing are addressed and shall be considered as part of the risk analysis:

- a) digital file creation for implants involves multiple steps and multiple parties
- b) specific acquisition and image data generation protocol, especially the resolution in 3 dimensions
- c) trying to improve or convert image data can lead to data corruption problems
- d) filtering, segmentation and reconstruction of medical image data (e. g. Digital Imaging and Communication in Medicine (DICOM)) into virtual 3D models and smoothing of these 3D models prior to additive manufacturing

NOTE:

This can for example include

- a) Correction of artifacts specific to imaging method used, e. g. beam-hardening and metal artifacts in CT;
- b) Special attention to landmarks like nerve or blood vessel openings in bone, which sometimes are not included in thresholding during segmentation;
- c) Image reconstruction due to poor image quality resulting from using low radiation dose (e. g. in case of young children, patients with healing wounds);
- d) Correction of 3D model when algorithms creating surfaces from masks smooth voxels touching at a thin point into an artificial gap or hole (step artifacts).
- e) Part location and part orientation on the build;
- f) Traceability of data to individual patients includes archiving build jobs.

6.6 Data processing: CAD to G-Code

6.6.1 The work flow of the transformation of a digital design into machine-executable data shall be based on standardized digital workflow.

6.6.2

The digital model shall be generated using any computer-aided design (CAD) software.

6.6.2.1

Model creation

- Use any validated CAD software to design the geometry
- Ensure the model is watertight

6.6.2.2

File preparation

- Export in STL/AMF/3MF format
- Verify units (mm vs inches) to avoid scaling errors

6.6.2.3

Validation

- Run a geometry check (mesh integrity, normals, manifold edges)
- Confirm compliance with additive manufacturing terminology and process category

6.6.2.4

Documentation

- Include metadata: process type (e.g., powder bed fusion), material, and build orientation

6.6.3

The model shall be defined within a right-handed Cartesian coordinate system, where the X- and Y-axes represent the build plane and the Z-axis denotes the build direction.

6.6.4 All geometric entities, features, and reference points shall be constructed relative to this coordinate system to ensure consistency across downstream data processing stages.

6.6.5 Units of measurement, orientation, and reference alignment shall be explicitly defined in accordance with [IS/ISO 80000](#) to ensure interoperability and dimensional accuracy during data exchange and manufacturing.

6.6.6 The digital model shall be validated for completeness and watertight geometry prior to export into an additive manufacturing file format for subsequent slicing and toolpath generation.

6.6.7

The G-Code arising from the slicing process should be a machine specific - executable numerical control instructions. The instructions (G-Code) shall define or contain information on the motion of the printing device in the X, Y, and Z axes including the machine parameters (Eg: Feed rate, extrusion rate, temperature, auxiliary functions).

6.6.8 The coordinate data shall correspond to either absolute or relative positioning modes consistent with the established build coordinate system.

6.6.9 Throughout this data conversion and transmission process, the integrity and traceability of digital information shall be maintained.

6.6.9.1 *Use standardized file formats:* Export models in STL, AMF, or 3MF to ensure compatibility and traceability across systems.

6.6.9.2 *Validate geometry integrity:* Run mesh checks to confirm watertightness, correct normals, and absence of non-manifold edges.

6.6.9.3 *Embed metadata in files:* Include process type, material, and build orientation within the file or accompanying documentation.

6.6.9.4 *Reference:* Records of design inputs, design outputs, model revisions, and validation activities shall be maintained to demonstrate conformity with the design requirements of this standard.

6.6.9.5 *Maintain version control:* Track revisions and conversions with timestamps and author details to preserve traceability.

6.6.9.6 *Secure transmission channels:* Use encrypted or authenticated file transfer methods to protect data integrity during transmission.

6.6.10 The reproducibility, dimensional accuracy, and conformity between the digital model and the manufactured component has to be ensured.

6.7 Design Verification**6.7.1**

Design verification shall confirm that:

- a) design inputs are correctly translated into outputs;
- b) the manufactured geometry conforms to the digital model within specified tolerances;
- c) mechanical and structural performance meet specified requirements;
- d) data processing steps do not compromise device performance.

6.7.2

Verification methods may include dimensional inspection, imaging-based evaluation, mechanical testing, and microstructural analysis.

6.7.3 *Finite Element Analysis (FEA):* the mechanical response of the implant under various loading conditions should be defined and analyzed in silico.

6.7.4 The FEA of the device as a whole or single part while preserving distinct geometric sections (Eg abutment and threaded regions).

7 AM PROCESS AND POST-PROCESSING**7.1 AM process: 3D Design****7.1.1** *Image and image processing*

- a) The digital models used to create 3D printed implants should match with required clinical application or to a patient's anatomy.
- b) **Implant Design:** The design for creating 3D construct to be used as an implant should be created using validated and licensed software.
- c) **Software for Image processing:** The software should be validated and approved to process medical digital images.

7.1.2 *Software Workflow:* The digital device design or the Computer Aided Design model shall be converted to a machine understandable file.

- a) The CAD model is divided into layers and converted into a new file

b) Complex designs can include additional support material to aid printing. However, the support material should also meet all the qualities of a biomaterial to be used clinically in a patient. The post printing processing to remove the support material should also be suitable for a biomaterial.

7.2 AM process: 3D printer

The 3D Printers used for manufacturing parts for use for clinical application should comply with the Compulsory Registration Order (RCRO), 2012 of BIS regarding sale, manufacture and import in India.

7.2.1 Printing: The object is printed using the design specifications included in the file.

7.2.2 Post-Processing: After printing is complete, one or more post-processing steps may be performed on the device or component. These may include cleaning to remove residual debris, controlled cooling (also called annealing), and/or additional steps such as drilling, cutting, polishing, and sterilization.

7.2.3 Process Validation and Verification: Some device or component characteristics can be checked individually after they are produced to make sure they will function properly and meet specifications. This is especially true of geometric features which can be checked quickly and nondestructively. When other functional features, such as mechanical strength, cannot be checked individually because the test could destroy the object or is impractical, manufacturers validate their processes prior to production. Process validation ensures that a manufacturing process will produce product that is within defined specifications, as long as specified processing parameters are monitored and controlled.

7.2.4 Testing: Device testing methods and results are submitted to the FDA to show that the device meets regulatory requirements and is reasonably safe and effective for its intended use. Each device or type of device has its own set of tests that may be based on FDA Guidance documents, international standards, or internal process controls. Devices manufactured by 3D printing are generally subject to the same regulatory requirements as more traditionally manufactured medical devices.

7.2.5 Requirements for post-processing for powder bed fusion. The following, but not limited to, factors related to the AM process and post-processing shall be considered as part of the risk analysis:

a) Mechanical properties - The mechanical properties of the implant are dependent on the feedstock and the manufacturing process. Mechanical properties for AM can be different from those from non-additive manufacturing processes. This can also result in different microstructure. The mechanical properties of the implant are also dependent on the location and orientation of parts within the build volume. Mechanical testing of implants manufactured using AM can have additional requirements for sample preparations, considerations for test setup and reporting requirements. It is important to understand gaps between requirements in existing implant standards and possible additional requirements for AM specific sample preparation, testing and reporting. If multiple different materials are used during the additive manufacturing process, like support materials in material jetting or in multimaterial implants, the interaction of these materials can have a significant influence on the properties of the manufactured implant.

b) Material properties - It is important to be aware that changes to the material properties can occur at all stages of the AM process and post-processing from feedstock to final output. For instance, oxygen intake or cross-contamination between different materials (e. g. in multimaterial builds) can occur during the AM process and affect the material properties (e. g. the physical and/or chemical properties).

c) Lattice structure - Lattice structure can for example be assessed with stereological evaluation, compression testing, X-ray diffraction and shear fatigue testing. It is important to be aware that complex features such as lattice structures can lead to unstable features at the boundary of the material.

d) Support structure - Some implant geometries might need structures to provide a base and anchor for the part during the AM process to manufacture the part as intended and ensure the intended properties of the finished implant. Supports are typically removed from the part prior to use. Considering the extent of removal as well as the method used for removal of the support structure can be critical during the risk analysis, since remaining support structures can for example affect the functionality of the finished implant.

e) Material-related post-processing - Common post-processing procedures to reach the intended material properties are thermal treatment, UV-light post-exposure or treatment with solutions of defined pH or salinity. Additively manufactured materials can behave differently compared to traditionally manufactured materials when exposed to common post-processing procedures.

f) Strut thickness - Strut thickness is the dimension of the single metal parts that form the framework of the porous layer.

7.3 AM process: Post printing process

a. The device created by additive manufacturing may undergo additional procedures like clearing the debris, annealing or controlled cooling, piercing, hurtful, smoothing or decontamination.

b. Process Validation - The device or its components created by additive manufacturing should be individually checked for its functional property and clinical specifications by non-destructive methods to ensure that the product is within defined specifications

7.4 Cleanliness and sterilization

7.4.1 The following factors related to cleanliness shall be considered as part of the risk analysis:

a) residual materials resulting from the AM process, such as:

1. remaining feedstock (e. g. sintered powder, resin, monomer);

2. remaining support structure;
 3. residues from machining, sand blasting, tumbling, colouring and other post-processes remaining inside the structures.
- b) complex structures (e. g. lattice/porous structure, internal channel, cavities), surface profiles and irregularities can add cleaning challenges due to the nature of AM processes. The stair case effect represents one surface profile which can occur in AM processes. Information on cleaning of orthopaedic implants can be found in . [IS 19227](#)

7.4.2 The following factors related to sterilization shall be considered as part of the risk analysis: suitability of the sterilization process, effects of sterilization on the product, exclusivity of the machine, and assurance on the cleanliness. Information on sterilization of orthopaedic implants can be found in . [IS 18076](#)

8 QUALIFICATION AND VALIDATION OF AM PROCESSES

AM processes used for production of surgical implants shall be qualified and validated. ISO/ASTM 52920 should be taken into account for the validation of AM processes. It is recommended to take into account any AM-technology-specific standards addressing the topics of installation, operation and performance qualification, if available.

9 TEST METHODS

9.1 Testing shall be done under ambient conditions.

9.1.1 Tribocorrosion

Tribocorrosion is an irreversible transformation of metal subsurface layers due to mechanical and chemical interactions caused by friction contact in a corrosive environment, in which wear or corrosion is not entirely responsible for material loss.

9.1.2 Roughness and decomposition microanalyses shall be determined in accordance with . [IS 3073](#) Sample preparation for biological evaluation shall be carried out in accordance with . [IS/ISO 10993](#)

9.1.3 Sterilisation

- a. Sterilisation should be included as a critical step in the manufacturing process.
- b. Select appropriate sterilization method suitable for the build material, biomaterial ink as well as the final 3D printed construct. (Eg for sterilization methods are Saturated steam, Low temperature steam with 2% formaldehyde, Gamma irradiation, Ethylene oxide gas, Hydrogen peroxide gas plasma, CO₂ under critical conditions)
- c. The sterilization methods should not affect the overall chemical nature of the product and its clinical functionality.
- d. Surface or bulk property analysis may be performed to ensure the characteristic of printed product.
- e. The sterilization process should ensure prevention of infections or rejections when the device comes into contact with the human body at the time of implantation.
- f. The new biomaterial without prior information on sterilization method is used, new sterilisation techniques may be adopted and the sterilization method has to be validated for the integrity and functionality of the 3D-printed devices.

9.2 Printing parameters

The parameters that can affect the dimensional stability of printed objects in Vat photopolymerization process is taken into consideration. Those parameters are as follows:

- a) layer thickness
- b) light intensity and exposure time
- c) temperature of vat and chamber

Printing parameters shall follow the manufacturers instructions for use. They shall be noted in the test report (if applicable).

9.3 Post printing analysis

The 3D printing parameters needs to be defined and documented Representative definite size structures may be designed and printed with to define the parameters: Definitive Parameters (The clear and final limits or criteria that define the printed part) needs to be defined and documented.

9.3.1 Design of alternating horizontal and vertical line filling can be designed with various dimensions, infill percentage, filament overlap, layer heights, etc.

9.3.2 Printing parameters or printer settings used for printing Design parameters needs to be documented.

9.3.3 physico chemical properties of each Definitive Parameter Print Output needs to be evaluated and physicochemical characterization as per the biomedical device evaluation needs to be performed.

9.3.4 Potentiodynamic test

Impedance study for Metal implants as per [IS 19491](#) The electrochemical impedance spectroscopy (EIS) test shall be conducted by immersing the test sample (working electrode) together with a KP-113 saturated calomel reference electrode and a PtP-201 platinum auxiliary electrode into 250 mL of Ringer's solution (NaCl 8.6 g/L, KCl 0.3 g/L, $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ 0.48 g/L) maintained at $37 \pm 1^\circ\text{C}$ and pH 7 ± 0.2 . The sample shall be stabilized for 120 minutes to determine its open-circuit potential (Eocp), after which EIS measurements shall be carried out using an AutoLab PGSTAT 302N system equipped with an FRA2 module. The impedance spectra shall be recorded in the frequency range of 10^4 Hz to 10^{-3} Hz with a sinusoidal voltage amplitude of 10 mV. The resulting data shall be fitted to an equivalent electrical circuit using the least squares method to calculate impedance, resistance, and capacitance values, and the results shall be presented as Nyquist and Bode plots for interpretation of surface layer properties.

9.3.5 Scratch test

The adhesion of layers to the titanium alloy substrates shall be evaluated using a scratch test performed with a Rockwell diamond cone indenter under gradually increasing normal load. The test shall be conducted in the loading force range of $F_c = 0.03\text{--}30$ N, with a loading rate of $v_s = 10$ N/min, a table speed of $v_t = 10$ mm/min, and a total scratch length of $l = 3$ mm. The critical load (L_c), representing the minimum normal force that causes loss of adhesion between the coating and the substrate, shall be determined based on recorded changes in acoustic emission, friction force, and friction coefficient, as well as by microscopic observation of the scratch track.

9.3.6 Microgap Analysis

The microgap inherently occurring or formed during the process of 3D printing may affect the mechanical stability and characteristic. Hence the porosity of the dispensed material in the Definitive Parameter Print Output shall be analysed and documented.

The printing parameters of end use implant device should be traced back to the Definitive Parameter Print Output such as:

- a. Design Parameters (Dimensions, Tolerances, Wall thickness, Hole size, Overhang angle)
- b. Printer Parameters (Nozzle diameter, Layer height, Print speed, Build volume, Curing parameters, material per product)
- c. Material Parameters (Material type, Nozzle temperature, Bed temperature, Cooling requirement)
- d. Slicing Parameters (Infill density, Infill pattern, Shells/Perimeters, Top-Bottom layers, Support material and settings, Adhesion type)
- e. Environmental Parameters (Ambient temperature, Printing temperature, Enclosure condition, Humidity)

9.4 Evaluation of 3D printed implants

The evaluation of additively manufactured medical device can be performed on the 3D printed implants, implant parts or the representative Definitive Parameter Print Outputs to collect information on the final performance of the device as a part or as a whole.

9.4.1 Micro-Computed Tomography

The 3D printed implants has to be analysed in a non-destructive method for the pore size, porosity, interconnectedness of pores and internal filling to ensure the expected structure and performance as an implant.

9.4.2 Scanning Electron Microscopy Analysis

- a. Extrude a filament or filament like construct using the biomaterial ink.
- b. Visually evaluate and record visual appearance like shape, distortions, extra material depositions and in fill patterns.
- c. Perform microscopy analysis to get information on dimensions of filament and calculate volume. Compare with the indented dimensions of theoretical values.
- d. 3D Print the construct using biomaterial ink.
- e. Take cross-sections of the samples by fracturing them in nitrogen.
- f. Use standard SEM processing and check the aspects such as porosity, pore size, pore connectivity, filament deposition and arrangement of the printed layers. Ref: Engineering Structures 216 (2020) 110752

9.4.3 Compressive Testing

- a) The compressive strength of the 3D printed implant to withstand the pressure without breaking or deforming, has to be evaluated by standard guidelines to ensure the desired function. (Eg. ASTM D 695)
- b) To measure compressive strength and compressive modulus of the biomaterial ink, the material may be casted or 3D printed in cylindrical or cubical shape to measure the stress and strain until failure.

9.4.4 Flexural Testing Procedure

Semi rigid biomaterial inks or flexible 3D printed implants The flexural strength and modulus of rigid and semi-rigid plastics were evaluated using a three-point bending test. Three samples each of extruded and 3D printed specimens were prepared and tested at a strain rate of approximately 1% min⁻¹ on a Universal Testing Machine (UTM), utilizing a

three-point bending setup. The specimens were assessed in their as-printed state, with representative images provided to illustrate the samples prior to testing.

9.4.4.1 Hardness testing procedure

The hardness of polymeric 3D printed specimens may be determined in accordance with validated test method. [IS 13360](#)
For metallic implants, hardness testing may be performed in accordance with [IS 1586](#)

9.4.5 Post-processing of Additively Manufactured Products

Additively manufactured products shall undergo post-processing, where applicable, to ensure conformity with specified requirements relating to mechanical properties, dimensional accuracy, surface characteristics, corrosion resistance, and biological performance.

Post-processing procedures shall be defined, validated, and controlled. The manufacturer shall establish documented acceptance criteria based on the intended use of the implant, material specification, design features (including porous or lattice structures), and applicable regulatory requirements. Post-processing may include one or more of the following, as appropriate:

a) Mechanical processes, such as machining, grinding, polishing, or abrasive blasting, to achieve specified dimensional tolerances and surface finish requirements. These processes shall not introduce surface damage, microcracks, or contamination. Functional porous or osseointegrative regions shall not be adversely altered unless specified by design.
b) Thermal processes, such as stress relieving, annealing, or hot isostatic pressing (HIP), to modify microstructure, reduce residual stresses, or improve mechanical performance. Time–temperature–pressure parameters shall be controlled and recorded. Where applicable, mechanical properties and microstructural characteristics shall be verified following treatment.

c) Chemical or electrochemical processes, such as etching, passivation, electropolishing, or coating, to improve surface finish, corrosion resistance, or biological response. Process parameters shall be controlled to prevent harmful residues. Coatings, where applied, shall meet specified requirements for thickness, adhesion, and uniformity.

Following post-processing, implants shall be cleaned using validated procedures to remove residual powders, process media, and contaminants. Cleanliness shall be verified by appropriate inspection or testing methods.

Verification shall demonstrate that post-processing achieves the intended functional outcomes without adversely affecting implant safety or performance. Records of post-processing parameters, inspection results, and validation data shall be maintained to ensure traceability.

9.5 The geometric capabilities such as accuracy, resolution etc. of the AM system shall be tested.

9.6

Star artefacts containing predefined geometric features may be used as reference specimens to assess the capability of non-destructive testing methods in detecting additive manufacturing defects such as porosity, lack of fusion, dimensional inaccuracies, and internal discontinuities.

The characteristics of the artefact, inspection method employed, and acceptance criteria shall be documented.

10 RISK ANALYSIS

The following subclauses identify factors which could have an impact on the characteristics related to safety and the intended performance of additively manufactured implants. However, they only identify factors which are either unique to additively manufactured implants or which require additional consideration for additively manufactured implants.

10.1 For each of the factors identified in 6.2 through 6.5 specific processes shall be specified and documented. [IS/ISO 14971](#) may be used for addressing the risk management.

10.2 A risk management process consists of multiple steps, of which only the risk analysis is addressed in this clause. Other steps of the risk management process, e. g. risk evaluation, heavily depend on the specific product in question and are therefore not addressed in this document.

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IS No.	IS Title
IS 19491:2026	Implants for Surgery Test Solutions and Environmental Conditions for Static and Dynamic Corrosion Tests on Implantable Materials and Medical Devices
IS 1586 :2025	Metallic materials Rockwell hardness testPart 1: Test method (Sixth Revision)
IS 19227 :2025	Breathing System Filters for Anaesthetic and Respiratory Use Part 1: Salt Test Method to Assess Filtration Performance
IS 17932 :2024	Biological Evaluation of Medical Devices Part 2 Framework for identification and quantification of potential degradation products
IS 17932 :2023	Biological evaluation of medical devices Part 1: Evaluation and testing within a risk management process (ISO 10993-1 : 2018, MOD)
IS 18076 :2023	Non-active surgical implants - General requirements
IS/ISO 80000 :2022	Quantities and Units Part 1 General
IS/ISO 14971 :2021	Medical devices - Application of risk management to medical devices First Revision
IS/ISO 10993 :2021	Biological Evaluation of Medical Devices Part 12 Sample Preparation and Reference Materials
IS/ISO 19227 :2018	Implants for Surgery - Cleanliness of Orthopedic Implants - General Requirements
IS/ISO 13485 :2016	Medical Devices — Quality Management Systems — Requirements for Regulatory Purposes (First Revision)
IS 13360 :1992	Plastics - Methods of testing: Part 5 mechanical properties section 13 determination of rockwell hardness
IS 3073 :1967	Assessment of surface roughness